

CHIP 2026 Hackathon — Team J5 (ITB)

Phase 0 Report: Foundations

ML-KEM-768 (FIPS 203) accelerator — golden model, FIPS 203 errata, official NIST ACVP known-answer tests, and an independent oracle cross-check

Status: DONE · Date (UTC): 2026-09-29 01:16

Branch: phase0-golden-model · HEAD: 7bc10cf · Not pushed to origin

Human approval: **NOT YET TICKED** (docs/results/result_phase0.md, Section 10) — this report does not authorize starting Phase 1.

Written for: the CHIPATON team (J5), as the record a human reviewer reads before ticking the Phase 0 approval box. Source of truth stays the repository (docs/results/result_phase0.md and docs/evidence/golden/); this PDF is a compiled snapshot of it, generated from those files.

1. Executive summary

Phase 0 built a pure-Python, standards-traced **golden model** of ML-KEM-768 (FIPS 203) in `tb/golden/`, read NIST's errata note for the standard, and verified the model three independent ways: 19 property tests, 80 official NIST ACVP known-answer test vectors, and 2000 randomized trials against an independent implementation (kyber-py). All three passed completely: **0 test failures, 0/80 KAT mismatches, 0/2000 cross-check mismatches**. No RTL, no FPGA numbers, and no timing/constant-time claim exist yet or are made in this phase.

Done-criterion (docs/ROADMAP.md)	Status
1. <code>check_params.py</code> passes; <code>k/eta1/eta2/du/dv</code> confirmed vs FIPS 203 Table 2	PASS
2. Golden model reproduces official ML-KEM-768 ACVP vectors	PASS
3. Errata findings recorded (evidence file + ADR)	PASS
4. Independent cross-check log (kyber-py oracle)	PASS
5. <code>result_phase0.md</code> passes <code>check_result.py</code> , human has approved it	PASS / pending tick

This is a technical DONE, not a go-ahead. Per CLAUDE.md's phase-gate rule, a human team member must read the evidence and tick the Approval checkbox in `docs/results/result_phase0.md` before Phase 1 (RTL for NTT/INTT) starts.

2. Scope and how this phase was run

Phase 0 followed a 6-prompt pack (`docs/prompts/phase0.md`), run one prompt at a time with a checkpoint after each:

- **Prompt 0 — Orientation.** Read CLAUDE.md, the project brief, the roadmap, open decisions, and `mlkem-guard`; ran `/phase-gate`; reported all criteria as MISSING (nothing existed yet). No file written.
- **Prompt 1 — FIPS 203 spec and errata.** Fetched the official FIPS 203 PDF and NIST's errata spreadsheet, hashed both, quoted and assessed every errata item, confirmed the ML-KEM-768 parameter row, and opened ADR 0003.
- **Prompt 2 — Constants and primitives.** Wrote `params.py` and `primitives.py` (NTT/INTT, sampling, encode/decode, compress/decompress, hash/XOF wrappers) with 19 property tests, all independent of any KAT vector.
- **Prompt 3 — K-PKE and ML-KEM.** Wrote `kpke.py` and `mlkem.py` (Algorithms 13-21: KeyGen/Encrypt/Decrypt, the deterministic internal KEM algorithms, the mandatory input checks, and the randomized public wrappers), with 4 end-to-end tests including a 500-trial round-trip.
- **Prompt 4 — Official NIST ACVP vectors.** Downloaded and sha256-verified two pinned ACVP-Server vector files, then compared the model against all five ML-KEM-768 test groups (80 cases total).
- **Prompt 5 — Independent oracle (kyber-py).** Installed kyber-py 1.2.0 in a throwaway virtualenv outside the repository and cross-checked 2000 pseudorandom trials against it.
- **Prompt 6 — Closeout.** Reran everything fresh, wrote `docs/results/result_phase0.md` from the template, validated it with `check_result.py` and `claim_lint.py` (0 errors both), updated only the roadmap status line and one CLAIMS_REGISTER.md row from proven evidence, and committed the work in 7 small commits on a feature branch — without pushing and without ticking the Approval box.

Hard rules that held throughout (CLAUDE.md): the FIPS 203 mathematics is locked (architecture-only innovation, `mlkem-guard`); every quantitative claim is labelled MEASURED or ESTIMATE with an evidence path; no test, tolerance, or comparison was weakened to force a pass; and no vector was invented — every KAT and hash came from a fetched, checksummed file.

3. FIPS 203 source and errata

Item	Value
FIPS 203 PDF	nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.203.pdf sha256 fe1f12f32a7e44ec9fdebbf400cda843a40b506dee676725234dc6f7923b6cac
Errata spreadsheet	csrc.nist.gov/.../fips-203-potential-updates.xlsx sha256 edf899c89762449f43d7713883caeefc2e4ae9ae98d5a76b339547db22cb3ac7
Access date	2026-09-28 UTC
Errata items found	2, both explicitly non-normative (“DO NOT introduce new technical requirements”)

#	Date identified	Location	Issue	Affects golden model?
1	2025-03-31	Appendix A	Zeta table includes $i=0$ (value 1) though Algorithms 9/10 only use $i=1..127$; correction just adds an explanatory sentence.	NO — zeta values are computed from BitRev7, not copied from the table; already consistent.
2	2025-10-17	Section 5.3, Alg. 15 line 7	Comment says “decode m from polynomial v ” but should say “ w ”; comment-only, algorithm body already uses w .	NO — golden model follows the algorithm body (w), not the erroneous comment.

Parameter confirmation (FIPS 203 Table 2, ML-KEM-768 row): $k=3$, $\eta_1=2$, $\eta_2=2$, $du=10$, $dv=4$, $n=256$, $q=3329$ — read directly from the PDF and found to match `tb/golden/params.py` exactly (no discrepancy, so no stop/report was triggered).

ADR 0003 (“FIPS 203 errata findings and golden-model handling”) is filed with Status: **Proposed**, not yet accepted by a named team decider. Until it is accepted, `docs/decisions/PENDING.md` item #9 stays open. This does not block any Phase 0 evidence above — it is a paperwork step for the team.

4. Golden model — files produced

Path	Purpose
tb/golden/params.py	Locked ML-KEM-768 constants (Q N ZETA K ETA1 ETA2 DU DV EK/DK/CT/SS_BYTES)
tb/golden/primitives.py	NTT / NTT ⁻¹ , MultiplyNTTs, BaseCaseMultiply, SampleNTT, SamplePolyCBD, ByteEncode/Decode, Compress/Decompress, hash/XOF wrappers
tb/golden/kpke.py	K-PKE.KeyGen / Encrypt / Decrypt (Algorithms 13-15)
tb/golden/mlkem.py	ML-KEM internal algorithms (16-18), §7.2/§7.3 input checks, randomized wrappers (19-21)
tb/golden/conftest.py	pytest import wiring for tb/golden/
tb/golden/tests/test_primitives.py	19 property tests (NTT round-trip, NTT-vs-schoolbook, 7-layer structure, encode/decode, compress bound, CBD range)
tb/golden/tests/test_mlkem.py	4 tests: 500-trial round-trip, bit-flip rejection, length checks, check_params.py re-run
tb/golden/fetch_acvp_vectors.py	Downloads + sha256-verifies the two pinned ACVP vector files
tb/golden/run_kat.py	Compares the model against the ML-KEM-768 ACVP groups
tb/golden/crosscheck_kyberpy.py	N=2000 deterministic cross-check against kyber-py

Every function in `primitives.py`, `kpke.py` and `mlkem.py` is commented with the FIPS 203 algorithm number and section it transcribes, and the NTT zeta table is computed from $\zeta=17$ via bit-reversal rather than pasted from memory. The model is explicitly **not** constant-time and makes no such claim; that is an RTL-level property for a later phase.

5. Test results

Suite	Result	Covers
pytest tb/golden/tests/	23/23 passed	19 primitive properties + 4 end-to-end KEM tests
check_params.py	12/12 OK	All locked constants match

5.1 Property tests (tb/golden/tests/test_primitives.py)

- (a) $\text{intt}(\text{ntt}(f)) == f$ for 50 random polynomials
- (b) NTT-domain multiplication == schoolbook negacyclic multiplication, 20 pairs
- (c)/(d) NTT and INTT each use exactly the 7 “incomplete NTT” layers (128,64,32,16,8,4,2 and reverse)
- (d) $\text{byte_decode}(\text{byte_encode}(f)) == f$ for $d \in \{1, 4, 10, 12\}$
- (e) Compress/Decompress: exact $\text{Compress}(\text{Decompress}(y))=y$ per FIPS 203 text, plus a rounding-error bound derived by the team (not copied from an explicit FIPS 203 formula — the standard only states this qualitatively)
- (f) SamplePolyCBD coefficients stay in $[-\eta, \eta] \bmod q$; wrong input length is rejected

5.2 End-to-end tests (tb/golden/tests/test_mlkem.py)

- 500 random-seed trials: Decaps(dk, Encaps(ek)) == K, every time
- 20 one-bit-flipped-ciphertext trials: deterministic $K\text{-bar} \neq K$ (implicit rejection), reproducible on repeat calls
- Encapsulation key, decapsulation key, ciphertext and shared-secret lengths all match
EK_BYTES/DK_BYTES/CT_BYTES/SS_BYTES
- check_params.py re-run as a subprocess inside the test suite

6. Official NIST ACVP known-answer tests

Source: github.com/usnistgov/ACVP-Server, pinned commit 975de31eb83d87039ec88934fdc47d8c312b892d. Both JSON vector files were downloaded and their sha256 verified against `kat_sources.md` before use; the JSON files themselves are not committed to the repository (`tb/vectors/acvp/*.json` is gitignored).

tgId	Function	Type	Total	Passed	Failed
2	keyGen	AFT	25	25	0
2	encapsulation	AFT	25	25	0
5	decapsulation	VAL	10	10	0
9	decapsulationKeyCheck	VAL	10	10	0
10	encapsulationKeyCheck	VAL	10	10	0
TOTAL			80	80	0

Of the 10 decapsulation cases, 5 are labelled “valid decapsulation” and 5 “modified ciphertext” (exercising the implicit-rejection branch); all 10 passed, including all 5 modified-ciphertext cases. Comparison method: exact byte match after hex decode, case-insensitive; for the two key-check groups, our boolean check function's result compared against the vector's `testPassed` field. **Zero tclds failed; nothing needed debugging.**

Honest correction: `kat_sources.md` assumed both ACVP files are NIST “sample sets” (`isSample: true`). As downloaded and hash-verified, the `ML-KEM-keyGen-FIPS203` file's own JSON says **`isSample: false`** (i.e. it is ACVP-Server's shipped full keyGen vector set for this revision); only the `encapDecap` file says true. Reported as found, not silently corrected.

Coverage caveat: only ML-KEM-512/-1024 groups were skipped; 80 ML-KEM-768 cases from one pinned commit is not exhaustive coverage of every input, failure path or invalid-key construction.

7. Independent oracle cross-check (kyber-py)

Item	Value
Oracle	kyber-py 1.2.0 (github.com/GiacomoPope/kyber-py), license MIT OR Apache-2.0
Install location	Throwaway virtualenv outside the repository (<code>~/cache/chip2026-oracle/venv</code>) — not added to <code>scripts/requirements-dev.txt</code> , not copied into the repo
Trials	N = 2000, fixed RNG seed 20260928 (reproducible)
Compared per trial	<code>keygen_internal</code> , <code>encaps_internal</code> (valid ciphertext), <code>decaps_internal</code> (1-bit-flipped ciphertext)
Result	2000/2000 trials fully matched at every stage; 0 mismatches

This is 2000 pseudorandom trials from one fixed seed — reproducible, but explicitly not treated as a proof (the raw log says so in its own text). It is run with the oracle venv's interpreter; run with the project's normal interpreter, the script stops immediately with a clear message instead of silently skipping the comparison.

8. Numbers (MEASURED)

Quantity	Value	Evidence
check_params.py	12/12 OK	cmd: check_params.py
pytest tb/golden	23/23 passed	cmd: pytest tb/golden/tests/
ACVP ML-KEM-768 KAT	80/80 passed	docs/evidence/golden/kat_mlkem768_2026-09-28.txt
kyber-py cross-check	2000/2000 matched	docs/evidence/golden/crosscheck_kyberpy_2026-09-28.txt
FIPS 203 errata items	2, both non-normative	docs/evidence/golden/fips203_errata_2026-09-28.md

9. Coverage and limits (what this phase does NOT prove)

- NIST sample/shipped vector sets only — 80 ML-KEM-768 cases from one pinned commit, not exhaustive.
- **ML-KEM-512 and ML-KEM-1024 are entirely untested** and not implemented (by design — this project targets ML-KEM-768 only).
- **This Python model is not constant-time** and no such claim is made; constant-time is an RTL-level property for a later, measured phase.
- kyber-py cross-check is 2000 trials from one fixed seed, not a proof.
- **No FPGA/Quartus numbers exist** — Phase 0 is Python-only; no ALM/register/M10K/DSP/Fmax or cycle-count claim is made or implied.
- §7.1 “key pair check” (seed-consistency + pairwise-consistency) is not implemented or tested — only the mandatory per-call §7.2/§7.3 checks are.
- Invalid-input rejection for our own check functions was exercised only through the 20 ACVP key-check cases; no additional fuzzing was done.
- ADR 0003 is Proposed, not yet accepted by a named team decider.

10. Deviations, decisions needed, and next steps

Deviations / failures: none. No test failed at any point; no tolerance, assertion, or comparison was weakened to force a pass. The one thing corrected publicly rather than hidden is the isSample discrepancy in Section 6 above.

Decisions needed from the team:

- Accept or revise ADR 0003 (errata handling) — needs a named decider and date.
- No other docs/decisions/PENDING.md item blocks Phase 0 closure; the rest target later phases.

Prerequisites before Phase 1 (NTT/INTT RTL, per docs/ROADMAP.md):

- A human ticks the Approval checkbox in docs/results/result_phase0.md, Section 10, after independently rerunning pytest and run_kat.py.
- ADR 0003 is accepted or revised.
- The branch is merged/pushed via scripts/setup_github.sh (not done by this session).
- Phase 1 itself then needs: cocotb bit-exact vs. golden on Verilator AND Icarus, clean verilator --lint-only -Wall and slang, evidence of constant cycle count, and a Quartus-compiled MEASURED resource/timing report.

11. Reproduce

```
# from repo root, branch phase0-golden-model
. scripts/env.sh

# 1. Golden model tests
python3 -m pytest tb/golden/tests/ -v

# 2. Locked parameters
python3 .claude/skills/mlkem-guard/scripts/check_params.py

# 3. Official NIST ACVP KAT comparison (downloads + sha256-verifies vectors first)
python3 tb/golden/fetch_acvp_vectors.py
python3 tb/golden/run_kat.py

# 4. Independent oracle cross-check (needs kyber-py==1.2.0 in a throwaway venv
#   OUTSIDE this repo, e.g. ~/.cache/chip2026-oracle/venv)
~/.cache/chip2026-oracle/venv/bin/python3 tb/golden/crosscheck_kyberpy.py

# 5. Validate the result artifact
python3 .claude/skills/phase-gate/scripts/check_result.py docs/results/result_phase0.md
python3 .claude/skills/proposal-claims/scripts/claim_lint.py docs/results docs/proposal
```

12. Appendix — commits on phase0-golden-model (not pushed)

```
7786950 docs: tambah prompt pack Fase 0, template/checker result-artifact, dan kat_sources.md
f42346e docs(fase0): temuan errata FIPS 203 dan ADR 0003 (Proposed)
241853c tb/golden: params.py + primitives.py (NTT, sampling, encode, compress) + 19 tes sifat
5c63a13 tb/golden: kpke.py + mlkem.py (Algoritma 13-21) + 4 tes end-to-end
edec57a tb/golden: bukti bit-exact vs 80 vektor ACVP resmi ML-KEM-768 (80/80 lulus)
bac5637 tb/golden: cross-check independen vs kyber-py 1.2.0, N=2000 (2000/2000 cocok)
7bc10cf docs(fase0): result_phase0.md (DONE, menunggu persetujuan manusia) + perbarui status
```

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